

DentalEase System Presentation

A Real-Time Dental Appointment and Clinic Management Platform

Prepared for system presentation and project defense

Project: DentalEase

Date: May 14, 2026

DentalEase is a web-based dental care management platform designed to make appointment booking, clinic scheduling, patient communication, and administrative verification easier, faster, and more organized.

System Overview

DentalEase provides a complete workflow for patients, clinics, and administrators.

- Patients can create accounts, book appointments, monitor appointment status, and receive realtime updates.
- Clinics can manage schedules, approve or decline appointments, receive instant notifications, and start telehealth sessions.
- Administrators can verify clinic and patient accounts and monitor system activity.

The system focuses on convenience, realtime communication, organized records, and a cleaner experience for dental care services.

Problem Addressed

Traditional dental appointment management often depends on manual communication, phone calls, delayed confirmations, and scattered records.

DentalEase addresses these common issues:

- Patients need a convenient way to request appointments online.
- Clinics need immediate visibility when a new appointment request arrives.
- Appointment status changes should not require manual page refreshes.
- Clinics need a structured schedule management page.
- Patients need simple access to appointment updates and dental records.
- Administrators need a controlled verification process for users.

User Roles

The system supports multiple user roles with different responsibilities.

Patient:

- Registers and waits for account approval.
- Books appointments with verified clinics.
- Receives appointment notifications and status updates.
- Views dashboard, appointments, and records.

Clinic:

- Registers and waits for admin verification.
- Receives new appointment requests.
- Approves or declines appointments.
- Manages schedules, patient records, and clinic profile.

Admin:

- Reviews account verification requests.
- Receives signup notifications.
- Manages and monitors platform activity.

Implemented Core Features

- Landing page and Get Started flow.
- Patient, clinic, and admin authentication.
- Signup and login pages with correct route behavior.
- Verification status workflow for patient and clinic accounts.
- Clinic profile management and clinic services support.
- Guest booking and authenticated patient booking.
- Patient dashboard, appointment page, and records page.
- Clinic dashboard, schedule management, records, and profile pages.
- Admin dashboard and verification notifications.
- Appointment approval and decline workflow.
- Telehealth/video consultation flow for supported appointments.
- Mobile responsive navigation and notification dropdown improvements.
- Pagination for long appointment and schedule lists.

Appointment Booking Workflow

The booking process is one of the central features of DentalEase.

Patient booking flow:

- Patient logs in to an approved account.
- Patient opens the appointment page.
- Patient selects clinic, service, appointment type, date, and time.
- Appointment is created with Pending status.
- Clinic receives a notification and sees the request in schedule management.

Clinic action flow:

- Clinic reviews pending appointment requests.
- Clinic can approve or decline the request.
- Patient receives a realtime notification.
- Patient appointment status updates automatically without manual refresh.

Appointment Status Handling

Implemented appointment statuses include:

Pending:

- Initial status after a patient books an appointment.
- Clinic can approve or decline this request.

Confirmed:

- Internal status after clinic approval.
- Patient-facing text displays as Waiting for Clinic for clearer communication.

Declined:

- Used when a clinic rejects an appointment request.
- Patient receives a notification and can choose another schedule.

Completed:

- Reserved for appointments that have already been finished.

Realtime Integration

DentalEase integrates Laravel Reverb for realtime communication.

Implemented realtime technologies:

- Laravel Reverb as the WebSocket server.
- Laravel Echo as the frontend WebSocket client.
- Pusher-compatible Reverb connection configuration.
- Private user channels for secure user-specific events.
- Realtime notification broadcast when app notifications are created.
- AppointmentChanged broadcast event for appointment create and update events.

Realtime behavior:

- Clinic receives updates when a patient books an appointment.
- Patient receives updates when clinic approves or declines.
- Dashboard and appointment sections refresh without a full page reload.
- A silent fallback sync supports reliability if a WebSocket connection is unstable.

Notification System

The notification system was implemented to support user awareness and realtime updates.

Implemented notification features:

- AppNotification model for persistent notification records.
- Notification creation for account signup, appointment submission, appointment approval, and appointment decline.
- Realtime notification broadcast through AppNotificationCreated event.
- Notification bell badge updates.
- Dropdown notification list updates.
- Mobile notification dropdown improved for smaller screens.
- Mark all notifications as read action.

The notification system helps reduce missed appointment requests and keeps both patient and clinic users informed.

Clinic Side Features

The clinic side is designed for appointment and patient management.

Implemented clinic features:

- Clinic dashboard with appointment statistics.
- Today schedule panel.
- Current queue panel.
- Performance summary including confirmed, pending, and total bookings.
- Schedule management page with search and schedule cards.
- Approve and decline appointment controls.
- Start Call button for confirmed telehealth appointments.
- Clinic records page grouped by patient activity.
- Clinic profile and service management.
- Schedule pagination set to 10 appointments per page.

Patient Side Features

The patient side is focused on easy access and appointment visibility.

Implemented patient features:

- Patient dashboard overview.
- Upcoming appointment summary.
- Active schedule section.
- Queue status section.
- Record summary section.
- Appointment booking modal.
- Clinic service selection based on chosen clinic.
- Calendar date picker for appointment date selection.
- Appointment list with status display.
- Patient records page.
- Appointment pagination set to 10 appointments per page.

Admin and Verification Features

The system includes administrative verification to improve platform trust and control.

Implemented admin-related features:

- Admin dashboard.
- Verification request handling.
- Notifications for new patient or clinic signup.
- User account approval workflow.
- Middleware checks for approved patient and clinic users.

Verification ensures that only approved clinics can receive bookings and only approved patients can book appointments.

Mobile UI Improvements

The mobile interface was refined to make the system easier to use on smaller screens.

Implemented mobile improvements:

- Responsive clinic and patient navigation.
- Mobile menu buttons for role dashboards.
- Notification dropdown converted into a full-width mobile panel.
- Notification list made scrollable on small screens.
- Appointment cards adjusted for mobile wrapping.
- Patient appointment controls adjusted to avoid overflow.

These improvements make important actions and notifications easier to access on phones.

Pagination Implementation

Pagination was added to improve performance and readability on pages with many appointments.

Implemented pagination:

- Clinic schedule management page displays 10 appointments per page.
- Patient appointment page displays 10 appointments per page.
- Existing summary counts remain based on the appropriate appointment set.
- Pagination links preserve query strings.

This prevents long pages, improves scanning, and keeps the user interface cleaner as the database grows.

Technical Stack

DentalEase was implemented using modern Laravel and frontend tooling.

Backend:

- Laravel PHP framework.
- Eloquent models and relationships.
- Blade templates.
- Middleware for roles and verification.
- MySQL database configuration.

Realtime:

- Laravel Reverb.
- Laravel Echo.
- Pusher JS client compatibility.

Frontend:

- Blade views.
- Tailwind CSS utility styling.
- Vite build pipeline.
- Responsive layouts for desktop and mobile.

Database and Model Structure

Key system data is organized through Laravel models and migrations.

Important models:

- User: stores patient, clinic, and admin users.
- Appointment: stores appointment booking information and status.
- AppNotification: stores notification records for users.
- VideoConsultation: stores telehealth consultation information.

Important relationships:

- Appointment belongs to patient user.
- Appointment belongs to clinic user.
- AppNotification belongs to user.
- AppNotification may belong to appointment.
- Appointment may have one video consultation.

Security and Access Control

Security and access control were considered in the system implementation.

Implemented controls:

- Authentication required for dashboards and protected pages.
- Role-based route protection for admin, clinic, and patient areas.
- Verification middleware for approved clinic and patient access.
- Private realtime channels for user-specific notifications.
- CSRF protection on forms.
- Server-side validation for booking forms.
- Clinic service validation when booking appointments.
- Authorization checks before clinics approve or decline appointments.

Testing and Validation

The project was checked using available Laravel and frontend validation tools.

Validation performed:

- PHP syntax checks for modified backend files.
- Laravel feature and unit test suite executed successfully.
- Vite production build executed successfully.
- Realtime event flow reviewed and updated.
- UI behavior refined for mobile notification dropdowns.
- Pagination behavior added to appointment-heavy pages.

These checks help confirm that the application remains stable after each improvement.

System Benefits

DentalEase provides practical benefits for patients, clinics, and administrators.

For patients:

- Easier appointment booking.
- Clear appointment status visibility.
- Realtime updates from clinics.
- Access to records and active schedules.

For clinics:

- Faster awareness of appointment requests.
- Better schedule organization.
- Easier approval and decline workflow.
- Centralized patient activity view.

For administrators:

- Better account control through verification.
- Improved monitoring of platform activity.

Conclusion

DentalEase is a complete dental appointment and clinic management system with realtime features, role-based workflows, account verification, responsive user interfaces, and organized appointment handling.

The system successfully integrates appointment booking, clinic schedule management, patient records, notifications, and realtime updates using Laravel Reverb.

With the implemented mobile UI improvements and pagination, the platform is better prepared for practical use, larger appointment volumes, and smoother patient-clinic communication.

DentalEase supports the goal of making dental care more accessible, organized, and responsive.